

MSF Diabetes Calculator

Paediatric Diabetic Ketoacidosis – Quick Start Guide

The MSF Diabetes Calculator provides a Paediatric Diabetic Ketoacidosis (DKA) feature to help you calculate the correct rate of intravenous fluids and insulin. This guide is designed to explain how to use this feature safely.



The MSF Diabetes Calculator is designed for use by prescribers. The output from this application must be checked and signed by a prescriber before clinical use. If you are unsure at any point, seek senior support.

Using the MSF Diabetes Calculator to manage Paediatric DKA

Step 1 – immediate management

- Assess the patient using a structured ABCDE approach and manage any life-threatening issues.
- Refer to MSF Paediatric Care section 9.2 diabetic ketoacidosis for guidance on diagnostic criteria and resuscitation of patients with DKA.

Step 2 – using the MSF Diabetes Calculator

- Collect the details needed to use the calculator:
 - Patient date of birth, sex, and weight
 - The operational centre and name of your MSF project
 - Patient blood glucose, urine or blood ketones, and (if available) blood gas results
 - The patient's Glasgow Coma Scale (GCS) score, if the patient is shocked or not, if the patient is on any respiratory support or not
 - If you do not have an infusion pump available, you will also need the drop factor of the fluid giving set you will be using
- Use a web browser on desktop, mobile or tablet to navigate to diabetes.msf.net.

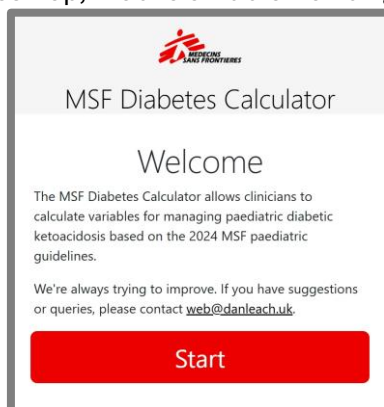


Figure 1 - MSF Diabetes Calculator start page

- Select **Start** then read the legal disclaimer and select **Agree and continue**.
Note: you can select the  icon next to some inputs to see more information about how data should be entered and how it is used.

- Complete the patient details form.

Note: provide an exact date of birth if possible – this is used to calculate the patient age and check the weight against a safety range. If you do not know the date of birth for your patient provide an age in years and months instead. Select the toggle button to show this option (see figure 2).

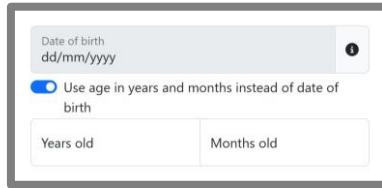


Figure 2 - Age in years and months alternative to date of birth



If an incorrect date of birth or age is provided, the weight safety check may be less effective. You should always measure the patient's weight carefully before using the application. The responsibility for using a correct weight lies with the user.

- Select **Continue** then complete the equipment availability form.



If an **infusion pump** is not available, you must select a drop factor which should be shown on the packaging of the giving set you are going to use. If the wrong drop factor is selected, the drops per minute values calculated will be incorrect.

IV DRIP RATES			
			
DROP FACTOR Number of drops (gtts) in one milliliter of solution	MICRODRIP Precise infusions; paediatrics/neonatal 60 gtts/ml	MACRODRIP Standard infusions; 20 gtts/ml	TRANSFUSION SET Blood product infusions with filter; 15 gtts/ml
*Always confirm gtts/ml (drop factor) as per manufacturer			

Figure 3 - drip rates used in MSF projects



If a **syringe pump** is not available, you must not give insulin via the intravenous route. Refer to MSF paediatric care for further guidance. The application currently will not provide insulin dosing guidance in this situation.

- Select **Continue** then complete the clinical details form.
- When you are ready select **Continue** again to generate guidance, you may need to wait a few moments depending on the speed of your network connection.
- The guidance should now be displayed. You must check the values are correct before clinical use. To help each calculated value is **highlighted** and can be clicked to view the working.

Step 3 – prescribing fluids and insulin

- The **MSF Paediatric DKA Fluid and Insulin Prescription Sheet** has been designed for use with the MSF Diabetes Calculator and can be downloaded from <https://diabetes.msf.net/msf-dka-fluid-insulin-prescription.pdf> if not already available locally.
- After checking the values are correct, transcribe these onto the prescription sheet and sign.
- Additional guidance is found on page two, including instructions on how to prepare fluids for the two-bag method.

Frequently asked questions

Can I use the MSF Diabetes Calculator offline?

The MSF Diabetes Calculator can be used offline, providing the web application has been loaded once. If you may need it when offline, visit the website ahead of time when you have a connection. Once the site is loaded the **Ready for offline use** label should appear in the footer (see figure 4).

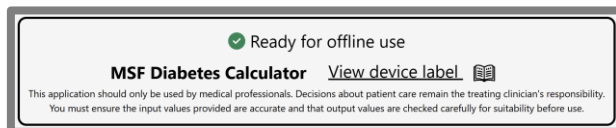


Figure 4 - Ready for offline use footer

You can also install the calculator as an app to your phone's home screen. See instructions on how to do this here: <https://www.installpwa.com/from/diabetes.msf.net>

Can I try out the MSF Diabetes Calculator?

You are welcome to test out the calculator. On the patient details form, please select **For testing or training purposes** rather than **For a real patient** to avoid confusing our audit data collection.

Can I give you a suggestion or let you know about a problem?

Yes, we'd be grateful! You can do this via the text box on the home page, or on the guidance page. Please leave your email so we can follow up if you're happy for us to do so.